INCH-POUND

MIL-R-23285B
AMENDMENT 7
6 February 1997
SUPERSEDING
AMENDMENT 6
7 July 1994

MILITARY SPECIFICATION

RESISTORS, VARIABLE, NONWIRE-WOUND
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-23285B, dated 13 September 1967, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

* Section 2, delete and substitute:

"2. APPLICABLE DOCUMENTS

"2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. The section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specifications, whether or not they are listed.

"2.2 Government documents.

"2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

"STANDARDS

"DEPARTMENT OF DEFENSE

"MIL-STD-202 - Test Methods for Electronic and Electrical Components Parts.
MIL-STD-1276 - Leads, Weldable, for Electronic Components Parts.

"HANDBOOKS

"DEPARTMENT OF DEFENSE

"Handbook H28 - Screw-Thread Standards for Federal Services.

"(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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"2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

"AMERICAN NATIONAL STANDARDS (ANS)

"J-STD-006 - Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications.

"(DoD activity may obtain copies of American National Standard J-STD-006 from the Defense Printing Service Detachment Office. Private sector and other Government agencies may purchase copies from the Institute for Interconnecting and Packaging Electronic Circuits (IPC), 7380 N. Lincoln, Lincolnwood, IL 60646.)

"AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

"ANSI/NCSL Z540-1 - Calibration Laboratory and Measuring and Test Equipment, General Requirements for.

"INTERNATIONAL ORGANIZATION for STANDARDS (ISO)

"ISO 10012-1 - Quality Assurance Requirements for Measuring Equipment, Part 1: Meteorological Confirmation System for Measuring Equipment.

"(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036-8002.)

"(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

"2.4 Order of precedence. In event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification shets, or MS sheets.), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

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3.2, at end of paragraph add: "Not applicable for style covered by MIL-R-23285/2."

Following 3.2, add:

"3.2.1 Production testing. Resistors furnished under MIL-R-23285/2 shall pass the production tests as specified."

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After 3.4.4.1, add:

"3.4.4.2 Solder dip (retinning) leads. Only the manufacturer may solder dip/retin the leads of product supplied to this specification provided the solder dip process has been approved by the qualifying activity. The manufacturer shall maintain a solder purity in accordance with table III-1, during the tinning process.

"3.4.4.2.1 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options:

a. When the original lead finish qualified was hot solder dip lead finish 52 of MIL-STD-1276. (NOTE: The 200-microinch maximum thickness is not applicable.) The manufacturer shall use the same solder dip process for retinning as is used in the original manufacture of the product.

b. When the lead original qualified was not hot solder dip finish 52 of MIL-STD-1276 as prescribed in 3.4.4.2.1a, approval for the process to be used for solder dip shall be based on the following test procedures:

(1) Thirty samples of any resistance value for each style and lead finish are subjected to the manufacturer's solder dip process. Following the solder dip process, the resistors are subjected to the dc resistance test (and other group A electricals). No defects are allowed.

(2) Ten of the 30 samples are then subjected to the solderability test. No defects are allowed.

(3) The remaining 20 samples are subjected to the resistance to soldering heat test followed by the moisture resistance test. No defects are allowed.

" TABLE III-1. Contamination limits.

Contamination	Tinning percent by weight 1/
Copper	0.750
Gold	0.500
Cadmium	0.010
Zinc	0.008
Aluminum	0.008
Antimony	0.500
Iron	0.020
Arsenic	0.030
Bismuth	0.250
Silver	0.750
Nickel	0.250

1/ This is a fixed percentage by weight of the solder.

"3.4.4.2.2 Solder dip retinning options. The manufacturer may solder dip/retin as follows:

a. Following the solder dip/retin process, total resistance test shall be repeated on the lot. Following the test, the manufacturer shall submit the lot to the solderability test as specified in 4.6.7.

b. As a corrective action if the lot fails the group A solderability test.

c. After group A inspection has been completed. Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 2, 100 percent screening tests shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 2 test. Following this test, the manufacturer shall submit the lot to the group A solderability test as specified in 4.6.7."

* 3.4.5, delete "QQ-S-561 or QQ-S-571" and substitute “J-STD-006”.

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PAGE 9

3.14, line 3, delete "±5 percent" and substitute "±2 percent".

PAGE 10

3.20, title, delete "Shock, medium impact" and substitute "Shock, specified pulse".

PAGE 11

* 4.1.1, delete and substitute:

"4.1.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with ANSI/NCSL Z540-1, ISO 10012-1, or equivalent system as approved by the qualifying activity."

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TABLE VI, group VII, column 1, delete "Shock, medium impact", and substitute "Shock, specified pulse".

PAGE 13

4.5, delete and substitute:

"4.5 Quality conformance inspection. Quality conformance inspection shall consist of group A and group B (see 4.5.1.2 and 4.5.1.3)."

4.5.1, delete and substitute:

"4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and group B inspections."

4.5.1.1, delete and substitute:

"4.5.1.1 Inspection lot. Inspection lot, as far as practicable, shall consist of all resistors of the same style and characteristic regardless of the resistance value produced in a period not to exceed 30 days, produced under essentially the same conditions, and offered for inspection at one time."

4.5.1.2, delete and substitute:

"4.5.1.2 Group A inspection. Group A inspection shall be as specified in table VII. The tests shall be conducted in the order shown."

TABLE VII, delete and substitute:

"TABLE VII. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	Sampling plan
Subgroup 1 Visual and mechanical examination	3.1, 3.3 to 3.4.8 inclusive 3.23 to 3.24.2 inclusive	4.6.1	See 4.5.1.2.1
Subgroup 2 Total resistance	3.5.1	4.6.2.1	See 4.5.1.2.2
Subgroup 3 Solderability	3.10	4.6.7	See 4.5.1.2.3

"

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Following TABLE VII, add:

"TABLE VII-1. Group A sampling plan.

Lot size	Subgroup 1 sample size	Subgroup 2 sample size
2 to 13	100 percent	100 percent
14 to 125	13	100 percent
126 to 150	13	125
151 to 280	20	125
281 to 500	29	125
501 to 1,200	34	125
1,201 to 3,200	42	125
3,201 to 10,000	50	192
10,001 to 35,000	60	294
35,001 to 150,000	74	294
150,001 to 500,000	90	345
500,001 and over	102	435

"

4.5.1.2.1, delete and substitute:

"4.5.1.2.1 Subgroup 1, sampling plan. A sample of parts from each inspection lot shall be randomly selected in accordance with table VII-1. If one or more defects are found, the lot shall be reworked or screened and defectives removed. After reworking or screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table VII-1. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification."

After 4.5.1.2.1, add:

"4.5.1.2.2 Subgroup 2, sampling plan. A sample of parts from each inspection lot shall be randomly selected in accordance with table VII-1. If one or more defects are found, the lot shall be reworked or screened and defectives removed. After reworking or screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table VII-1. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

"4.5.1.2.3 Subgroup 3.

"4.5.1.2.3.1 Sampling plan. A sample of five parts from each inspection lot shall be selected and subjected to the subgroup 3 solderability test. If there are one or more defects, the lot shall be considered to have failed.

"4.5.1.2.3.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot:

a. Each production lot that was used to form the failed lot shall be individually submitted to the solderability test as required in 4.6.7. Production lot samples that pass the solderability test can be reworked only if submitted to the solder dip procedure in 4.5.1.2.3.2b.

b. The manufacturer submits the failed lot to a 100 percent solder dip using an approved solder dip process in accordance with 3.4.4.2. Following the solder dip, the total resistance test shall be repeated on 100 percent of the lot. Lot acceptance criteria for total resistance shall be as for subgroup 1 tests. A sample of five parts shall be randomly selected and subjected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot may be reworked a second time and retested. If the lot fails this, the second rework, the lot shall be considered rejected and shall not be furnished against the requirements of this specification.

"4.5.1.2.3.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract."

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TABLE VIII, delete and substitute:

"TABLE VIII. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	Sampling plan
Resistance taper	3.5.2	4.6.2.2	See 4.5.1.3.1
Minimum resistance	3.5.3	4.6.2.3	
Torque	3.6	4.6.3	
Dielectric withstanding voltage	3.7	4.6.4	
Contact-resistance variation	3.8	4.6.5	

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4.5.1.3.1, delete and substitute:

"4.5.1.3.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table IX. If one or more defects are found, the lot shall be reworked or screened and defectives removed. After reworking or screening and removal of the defectives, a new sample of parts shall be randomly selected in accordance with table IX. If one or more defects are found in the second sample, the lot shall not be supplied to this specification."

TABLE IX, delete and substitute:

"TABLE IX. Group B sampling plan.

Lot size	Sample size
2 to 5	100 percent
6 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1,200	19
1,201 to 3,200	23
3,201 to 10,000	29
10,001 to 35,000	35
35,001 and over	40

4.5.1.4, delete.

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4.5.1.4.1, 4.5.1.4.2, and 4.5.1.4.3, delete.

* 4.5.2, delete and substitute:

"4.5.2 Inspection of preparation for delivery. Sample packages and packs shall conform with the requirements in section 5 of this specification."

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4.5.3, delete and substitute:

"4.5.3 Retention of qualification. Every 6 months, the manufacturer shall compile a summary of the results of quality conformance inspections in the form of a retention of qualification report, and forward it to the qualifying activity within 30 days from the end of the reporting period as the basis of continued qualification approval. In addition, the manufacturer shall immediately notify the qualified activity whenever the group B inspection data indicates failure of the qualified product to meet the requirements of the specification. Continuation shall be based on the evidence that over the 6-month period, the following has been met:

a. The manufacturer has not modified the design of the item.

b. The specification requirements for the item have not been amended so far as to affect the character of the item.

c. Lot rejection for group A inspection does not exceed the group A sampling plan.

d. The requirements for group B inspection are met.

When group B requirements are not met and the manufacturer has taken corrective action satisfactory to the government, group B retesting shall be instituted. A summary of the retesting shall be forwarded to the qualifying activity within 30 days after completion of the retest. All reports are to be certified by a responsible company official and the government inspector."

4.6.2, delete and substitute:

"4.6.2 DC resistance (see 3.5). Resistors shall be tested in accordance with method 303 of MIL-STD-202. The following details shall apply:

a. Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same style, model, or if it can be shown that the performance of the equipment is equivalent or better.

b. Limit of error of measuring apparatus: Shall not exceed ±0.5 percent for qualification inspection nor ±1 percent for quality conformance inspection.

c. Voltage: As specified in table X. Voltages shall be those actually impressed across the resistors. The voltage selected shall be used for all tests and all resistors having the same nominal total resistance value shall be measured with the same test voltage."

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TABLE XI, delete and substitute:

"TABLE XI. Resistance-temperature coefficient.

Sequence 1/	Temperature °C ±2°C
1	Room temperature 2/
2	-15
3	-65
4	Room temperature 2/
5	+65
6	+125
7	+150
8	+175

1/ At the option of the manufacturer, the reverse sequence may be as follows:
a. Room temperature.
b. +175 ±2°C.
c. Room temperature.
d. -65 ±2°C.

2/ This temperature shall be considered the reference temperature for each of the succeeding temperatures."

PAGE 21

4.6.9.2, under column "Bushings L and T", delete "0 cycle". Under column "Bushings S and N", delete "0 cycle" and substitute "1,000 cycles".

PAGE 24

4.6.17, delete and substitute:

"4.6.17 Shock, specified pulse (see 3.20). Resistors shall be tested in accordance with method 213, test condition I, MIL-STD-202 (100G, 6 ms)."

PAGE 27

* 5. and 5.1, delete and substitute:

"5. PACKAGING.

"5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department*s System Command. Packaging data retrieval is available from the managing Military Department*s or Defense Agency*s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity."

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* 6. delete and substitute:

"6. NOTES.

"(This section contains information of a general or explanatory nature which may be helpful: but is not mandatory.)"

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6.1.1, delete.

6.2, line 9, delete "Electronics Command" and substitute "U.S. Research Laboratory"; line 11, delete "(DESC-E)" and substitute "(DSCC-VQP)"; delete "Engineering Standardization Directorate"; last line, delete "45401" and substitute "43216-5000".

FIGURE 6, delete and substitute:

"

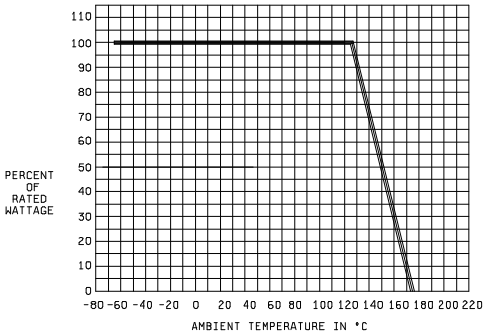

FIGURE 6. Derating curve for high ambient temperatures."

* After 6.8.3, add:

"6.8.4 Shaft length substitution data. Shaft length symbols "C" and "A" of MIL-R-23285A(NAVY) are replaced as follows under revision B of this specification:

Shaft length symbol of MIL-R-23285A(NAVY)	Revision B shaft length symbol
C (5/8 inch)	A (5/8 inch)
A (3/8 inch)	L (3/8 inch) "

6.10, title, delete "Shock, medium impact and vibration, high frequency." and substitute "Shock, specified pulse."

* 6.12, delete and substitute:

"6.12 Copies of Provisions Governing SD-6 may be obtained from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.

"6.12.1 Subject term (key word) listing.

Film resistance element
Bushing
Shaft
Linear taper

* 6.13, delete and substitute:

"6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes."

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30.2, bushing types, delete and substitute:

"Bushing types:

N or S will qualify L or T when lot 5 of TABLE XII is submitted with an N or S (lots 1 through 4) submission. Qualification of resistors with standard (N) type bushing shall qualify resistors with sealed (S) type bushing. Conversely, qualification of resistors with sealed (S) type bushing shall qualify resistors with standard (N) type bushing. Qualification of resistors with locking (L) type bushing shall qualify resistors with sealed (T) type bushing. Conversely, qualification of resistors with sealed (T) type bushing shall qualify resistors with locking (L) type bushings."

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Navy - EC
Air Force - 85

Review activities:
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Preparing activity:
DLA - CC

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